Correlators of boosted resonances

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1 Motivation

Measuring EECs of vector bosons is highly appealing. Potential applications include:

- Search for boosted narrow resonances decaying hadronically (as a bump hunt in the EEC spectrum). (Are we sensitive to Higgs??).
- Calibration/proof-of-principle for EEC top mass measurement. The actual top mass measurement is a separate work, this is just a first demonstration of the technique.
- Measurement of transverse/longitudinal polarization fraction of high-pT Z and W
- Measurement of the full production spin-density matrix for high-pT Z and W bosons

2 Background

The two-point energy correlator (EEC) of a boosted, hadronically decaying boson looks very different from that of a QCD jet, whose collinear spectrum is smooth and approximately scale invariant [1]. A boson of boost $\gamma = E_V/m_V$ instead produces a decay peak at $z' \simeq 1/\gamma^2$, with $z' = (1 - \cos\theta')/2$, which at central rapidity corresponds to $\Delta R \simeq 2m_V/p_T$ [1]. The width of the peak is set mainly by QCD radiation from the quarks rather than by Γ_V/m_V, and well below the peak the spectrum follows that of a quark jet [2].

2.1 Polarization and production

In the narrow-width approximation, the correlator factorizes into a production density matrix and a decay matrix [1, 2]. The opening-angle spectrum depends on production only through the longitudinal fraction ρ_{00} [1]. Using NNLO production for $pp \to V + X$ and decay correlators at NNLO with NNLL collinear and N^{4}LL back-to-back resummation, Gao et al. find that the peak height at $\gamma = 4$ differs by about 70% between purely longitudinal and purely transverse bosons. For $p_T > 100$ GeV at $y_V = 0$, the polarized spectrum is suppressed by up to about 10% across the peak relative to an unpolarized treatment, because high-p_T production is dominantly transverse [1]. The off-diagonal entries require measuring the detector orientation relative to the production plane [1]: $|\Delta h| = 2$ interference appears in the azimuthal angle, while $|\Delta h| = 1$ interference requires the azimuth and the angular separation jointly [2]. One-point energy and charge correlators together determine all eight real parameters of the density matrix [1]. Two experimental cautions come from Ref. [2]: boosts along the LHC beam axis modify the azimuthal distributions, an effect expected to be mild but not yet assessed, and in a lepton-collider study D_2 selections preferentially reject transverse bosons, homogenizing the longitudinal and transverse one-point spectra.

Two neighboring results extend the toolkit. The one-point correlator of a vector current is fixed by a single parameter $a_\mathcal{E}$, bounded by unitarity to $-1/2 \le a_\mathcal{E} \le 1$; its N^{3}LO value

at $q^2 = m_Z^2$ is -0.42, in good agreement with values extracted indirectly from LEP data [3]. Flavor flux correlators generalize the program to flavor quantum numbers, with hadronization corrections to the b and c cases suppressed by $\Lambda_{\mathrm{QCD}}/m_Q$ [4].

2.2 Searches

The same peak motivates resonance searches. For a narrow resonance coupled linearly to a QCD operator, the peak position encodes the mass and the spectral shape encodes the spin and polarization, within a space of shapes bounded by unitarity and energy positivity [5]. A projection for a light hadrophilic Z' in $pp \to Z'\gamma$, with statistical uncertainties only, finds that the inclusive EEC reaches sensitivity comparable to a recast of a CMS soft-drop mass search [5]. A separate recast of the CMS dijet correlator measurement also gives competitive limits, although its signal is at particle level and detector effects may degrade the reach [5]. In a Monte Carlo study of jets with $p_T > 1$ TeV recoiling against $Z \to \ell\ell$, hadronic W/Z decays appear as a local excess of about 7% even though they make up only about 1% of the jets, with no reconstruction or tagging; that selection is better suited to the HL-LHC than to existing datasets [1]. The peak position also calibrates the angular scale, making m_V a standard candle inside jets, including the polarized W inside boosted top jets [1].

3 Questions and Goals

In the short term, we want to understand how we can use these ideas to do a CMS measurement. A few important lines of questioning:

"Untagged" resonance search

- What sensitivity might we expect in an "untagged"-style search for a narrow resonance in Run 3 data?
- Which dataset(s) would we want to use?
  - Z+jet (tagged by Z$\to \ell\ell$)? Potential advantage: polarization of the other Z from leptonic decay?
  - W+jet (tagged by W$\to \ell\nu$)? Potential advantage: polarization of the other W from leptonic decay?
  - Photon+jet (tagged by photon)? Potential advantage: larger cross-section. Photon cannot be longitudinally polarized, I assume this has some implications for the polarization of the Z?
  - Dijet (with exactly two jets and momentum balance)? Potential advantage: large cross-section, but also large QCD background.
  - QCD multijet (without momentum balance)? Potential advantage: largest cross-section, but also largest QCD background.
  - Something else?
- What do various SM processes look like?
  - Can we separate W and Z at all?
  - Can we see the Higgs? If not, can we set limits? Would they be competitive?
  - What does the top look like? Do we learn anything interesting?

- Do we get any extra sensitivity from observables beyond two-point correlator? E.g., E3C, E1C, etc. Including non-standard formulations thereof? The paper mentions E^2E^2 for example. Or with non-standard angles as in the one-point from the paper...
- Would we get a boost in sensitivity from some additional cuts?
- Is heavy flavor interesting? Can we see heavy mesons?
- Is the background really as clean as we hope it is?
- How bad is the jet energy uncertainty? Is that the dominant systematic? What other systematics could be important?
- Would our limits actually be competitive?
- What jet radius to use? I assume AK8
- Use tracks or PF candidates?
- Useful to discriminate signals by spin/polarization? Or just an unpolarized bump hunt? Separate limits for vector vs scalar? What would an axial-vector signal look like? Or a pseudoscalar?

Polarization measurements and spin density matrix

- How good of a job do we need to do at tagging? Can we do it untagged? Is there any way to do good tagging that doesn't badly sculpt the distributions?
- Is there a useful shape difference between longitudinal and transverse bosons? Or is it dominantly the peak height?
- Are the one-point correlators actually practical? What does the background do to them?
- Would the three-point correlator be useful/interesting? Or some related things, maybe non-standard ones...
- Is there an interesting bell inequality story here? Can we see interference effects?
- Again, which dataset(s) would we want to use?
- Charge correlators? Are they useful? How badly does hadronization etc. ruin the picture?
- How large are the spin analyzer coefficients?
- What is the other literature? Are we competitive?
- What jet radius to use? I assume AK8
- Use tracks or PF candidates?
- What is the best approach for getting the full matrix? Minimal set of most useful correlators? Compare one-point to two-point to three-point?
- Link to EFT?

4 Tasks

We will refine the procedure as we learn, so each step below ends with a check-in to review the results and revise the later steps. Quantitative metrics for comparing channels, selections, and observables will be defined at these check-ins, once the procedure is better understood.

1. **Literature.** Read more literature to get a sense of what has already been done, and fix the mass and p_T ranges of interest.
   - What are the current limits on narrow resonances in the mass range of interest? Start from the references in Ref. [5].
   - What are the current uncertainties on the W and Z longitudinal fractions, as a function of p_T? Start from the references in Ref. [1].
   - What are the current uncertainties on the full spin density matrix, as a function of p_T?
   - *Check-in.*
2. **Reproduce and explore.** Set up particle-level correlator code, starting with only the Z($\ell\ell$) channel, and scale out from there.
   - Required samples: Z($\ell\ell$)+jet, WZ, and ZZ, where the diboson samples have Z$\to \ell\ell$ and a hadronic W or Z. All of these are needed, because the published result is built from their sum.
   - Two-point correlator: reproduce the Z($\ell\ell$)+jet result in Fig. 8 of Ref. [1].
   - One-point correlator: explore it ourselves in the same samples. We are not targeting an exact reproduction of published one-point results, which use different samples [2].
   - *Check-in.*
3. **Samples and data formats.**
   - Survey which of the samples in Table 1 are already available centrally, and use them when possible.
   - We will probably need to skim a custom NANO format from MiniAOD to get at the jet constituents. Check whether everything we need is in MiniAOD, or if we need full AOD.
   - *Check-in.*
4. **Polarization feasibility.** Using longitudinal and transverse samples, determine how sensitive the correlators are to the longitudinal fraction.
   - Check empirically how the choice of jet radius affects longitudinal and transverse bosons.
   - *Check-in:* decide whether the SM polarization measurement remains a goal.
5. **Search exploration.** Add the remaining samples a few at a time, and experiment with different channels, jet radii, selections, cuts, and observables to see what is most promising, using a jet-mass analysis as a baseline.
   - *Check-in.*
6. **Detector-level studies.** Add detector simulation, systematic uncertainties, a background estimation method, and a statistical model that accounts for correlations between bins.

- *Check-in.*

7. **Proposal.** Based on these studies, set up a minimal proposal for a measurement and/or search, including a sensitivity estimate. Rediscovering the W and Z is the natural first milestone.

Samples Table 1 lists the samples we expect to need. Take care in sample generation to keep track of the polarization state of every boson we simulate.

Table 1: Planned simulated samples.

Sample	Decays and variants	Notes
Z' resonances	Small number of mass points to start	
Scalar resonances	Small number of mass points to start	
Higgs	Various decay channels	Can we see decay to light quarks?
Z+jet	$Z\to \ell\ell$ and $Z\to q\bar{q}$, separately	$Z\to \ell\ell$ required for step 2
W+jet	$W\to \ell\nu$ and $W\to q\bar{q}'$, separately	
$V\gamma$	Leptonic and hadronic V decays, separately	
Photon+jet		
Diboson	All permutations of leptonic and hadronic decays	WZ and ZZ with $Z\to \ell\ell$ and a hadronic W or Z required for step 2
Multiboson	All permutations of leptonic and hadronic decays	
QCD multijet		
Single-top	All channels	
ttbar		

5 Resources

List required resources, dependencies, or tools.

6 Risks and Mitigations

Identify potential risks and how they will be addressed.

7 Success Criteria

Define what constitutes successful completion.

References

[1] Anjie Gao, Kyle Lee, and Xiaoyuan Zhang. *Precision Studies of Boosted Polarized Vector Bosons with Energy Correlators*. 2026. arXiv: 2610.38688 [hep-ph].

[2] Lorenzo Ricci and Marc Riembau. "Energy Correlators of Hadronically Decaying Electroweak Bosons". In: *Physical Review D* 106.11 (2022), p. 114010. DOI: 10.1103/PhysRevD.106.114010. arXiv: 2207.03511 [hep-ph].

[3] Marc Riembau and Minho Son. "Flow between Extremal One-Point Energy Correlators in QCD". In: *Physical Review Letters* 136.10 (2026), p. 101901. DOI: 10.1103/lc5m-kwgv. arXiv: 2509.02669 [hep-ph].

[4] Kyle Lee and Ian Moult. *Flavor Flux Correlators, Forward-Backward Asymmetries, and Anomalies*. 2026. arXiv: 2609.13366 [hep-ph].

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